Dorey's Resource Book

Topic 3 – Right-angled triangles

Contents

Episode 1. Pythagoras' theorem

Episode 2. Pythagoras in practical problems

Episode 3. The trigonometric ratios

Episode 4. Finding a side: unknown on top

Episode 5. Finding a side: unknown on the bottom

Episode 6. Degrees and minutes

Episode 7. Finding an angle

Episode 8. Angles of elevation and depression

Episode 9. True and compass bearings

Episode 10. Bearing problems

1 Pythagoras' theorem

ANCHOR QUESTION

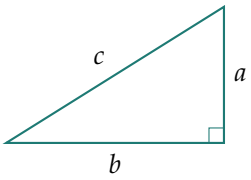
A right-angled triangle has shorter sides 9 and 12. Find the length of the hypotenuse.

SYLLABUS MST-12-S1-04

- Apply Pythagoras' theorem to solve practical problems in two dimensions

NOTES

- The **hypotenuse** is the longest side of a right-angled triangle. It is opposite the right angle.
- **Pythagoras' theorem:** $c^2 = a^2 + b^2$, where c is the hypotenuse.
- To find the hypotenuse, add the squares: $c = \sqrt{a^2 + b^2}$.
- To find a shorter side, subtract the squares: $a = \sqrt{c^2 - b^2}$.
- Write all sides in the same units first. Round only at the end.



WORKED EXAMPLE

A right-angled triangle has shorter sides 8 and 15. Find the length of the hypotenuse, x.

$x^2 = 8^2 + 15^2$ hypotenuse: add the squares

$x^2 = 64 + 225 = 289$

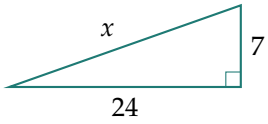
$x = \sqrt{289} = 17$ square root

Tier 1 · Finding the hypotenuse

1. The shorter sides of a right-angled triangle are 3 and 4. Find the hypotenuse.

2. The shorter sides of a right-angled triangle are 5 and 12. Find the hypotenuse.

3. Find x.



Tier 2 · Rounding

4. The shorter sides of a right-angled triangle are 4 and 9. Find the hypotenuse, correct to two decimal places.

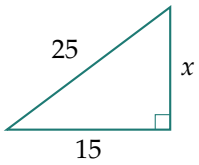
5. The shorter sides of a right-angled triangle are 2.5 and 6. Find the hypotenuse.

6. The shorter sides of a right-angled triangle are 11 and 14. Find the hypotenuse, correct to one decimal place.

Tier 3 · Finding a shorter side

7. A right-angled triangle has a hypotenuse of 10 and one shorter side of 8. Find the other shorter side.

8. Find x.



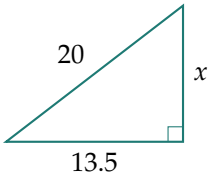
9. A right-angled triangle has a hypotenuse of 17 and one shorter side of 8. Find the other shorter side.

Tier 4 · Shorter side with rounding

10. A right-angled triangle has a hypotenuse of 12 and one shorter side of 7. Find the other shorter side, correct to two decimal places.

11. A right-angled triangle has a hypotenuse of 8.5 and one shorter side of 4.2. Find the other shorter side, correct to two decimal places.

12. Find x, correct to two decimal places.



Tier 5 · Mixed units

13. The shorter sides of a right-angled triangle are 60 cm and 0.8 m. Find the hypotenuse in metres.

14. A right-angled triangle has a hypotenuse of 2.6 m and a shorter side of 100 cm. Find the other side in metres.

15. The shorter sides of a right-angled triangle are 1.5 km and 800 m. Find the hypotenuse in kilometres.

Tier 6 · Two triangles

16. In triangle ABC, D is a point on BC and AD is perpendicular to BC. $AB = 13$, $BD = 5$ and $DC = 16$. Find AD, then find AC.

17. In triangle ABC, AD is perpendicular to BC. $AB = 10$, $BD = 6$ and $DC = 6$. Find AC.

18. A square has sides of 7 cm. Find the length of its diagonal, correct to two decimal places.

Conceptual

19. Use Pythagoras' theorem to explain why the hypotenuse is always the longest side.

20. A triangle has a hypotenuse of 10 and a shorter side of 6. Mia finds the other side as $\sqrt{100 + 36} = 11.66$. Explain her mistake and find the correct answer.

21. A square has a diagonal of 10 cm. Find the side length, correct to two decimal places.

22. Explain why the sides must be in the same units before using Pythagoras' theorem.

23. A composite shape is made of a right-angled triangle with shorter sides 9 cm and 12 cm, and a semicircle drawn on the hypotenuse.

(a) Find the diameter of the semicircle. (2 marks)

(b) Find the perimeter of the composite shape, to the nearest centimetre. (2 marks)

Revision

24. A spinner with 5 equal sections numbered 1 to 5 is spun and a coin is tossed.

(a) How many outcomes are there? (1 mark)

(b) Find $P(\text{head and an even number})$. (1 mark)

(c) Find the probability of NOT getting a head and an even number. (1 mark)

(d) Find the expected number of these results in 300 goes. (2 marks)

25. A garden bed is a trapezium with parallel sides of 6 m and 9 m, which are 4 m apart.

(a) Find the area of the garden bed. (2 marks)

(b) Mulch is spread 0.1 m deep. Find the volume of mulch needed. (1 mark)

(c) Mulch costs \$65 per m^3, plus a \$45 delivery fee. Find the total cost. (2 marks)

26. A tap fills a 9000 L tank at a rate of 15 L/min.

(a) How many hours does it take to fill the tank? (2 marks)

(b) Convert 15 L/min to L/h. (1 mark)

(c) Water costs \$3.20 per kilolitre. Find the cost of the water to fill the tank. (2 marks)

2 Pythagoras in practical problems

ANCHOR QUESTION

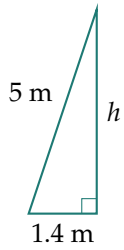
A 5 m ladder leans against a wall with its foot 1.4 *m from the wall. How high up the wall does it reach?*

SYLLABUS MST-12-S1-04

- Apply Pythagoras' theorem to solve practical problems in two dimensions
- Solve practical problems involving Pythagoras' theorem, the trigonometry of right-angled triangles, angles of elevation and depression, and true bearings and compass bearings

NOTES

- Draw a right-angled triangle for the situation and label the lengths you know.
- The hypotenuse is the side opposite the right angle: the ladder, the diagonal, the kite string, the sloping ramp.
- Walking due north then due east (or any two perpendicular directions) makes a right angle.
- Answer the question asked: a saving or a difference may need one more step after Pythagoras.
- State units and round as asked.



WORKED EXAMPLE

A 6 m ladder leans against a wall. Its foot is 2 m from the wall. How high up the wall does it reach, correct to two decimal places?

The ladder is the hypotenuse. — opposite the right angle
$h^2 = 6^2 - 2^2 = 32$ — shorter side: subtract
$h = \sqrt{32} = 5.656\ldots \approx 5.66$ m — round at the end

Tier 1 · Ladders and walls

27. A 3.5 m ladder leans against a wall with its foot 1 m from the wall. How high up the wall does it reach, correct to two decimal places?

28. A ladder reaches 4 m up a wall. Its foot is 1.2 m from the wall. How long is the ladder, correct to two decimal places?

29. A 10 m ladder reaches 8 m up a wall. How far is its foot from the wall?

Tier 2 · Diagonals

30. A rectangular phone screen is 15 cm by 7 cm. Find its diagonal, correct to one decimal place.

31. A gate is 1.8 m wide and 1.2 m high. A diagonal brace is fitted across it. Find the length of the brace, correct to two decimal places.

32. A TV is sold as having a 65-inch diagonal. Its width is 57 inches. Find its height, correct to the nearest inch.

Tier 3 · Directions

33. Ella walks 600 m due south, then 800 m due west. How far is she from her start, in a straight line?

34. A boat sails 12 km due east, then 5 km due north. How far is it from its starting point?

35. A plane flies 230 km due north, then 95 km due east. How far is it from its starting point, correct to the nearest kilometre?

Tier 4 · Comparing distances

36. A rectangular field is 120 m by 50 m. How much shorter is the walk along the diagonal than along two sides?

37. A square courtyard has sides of 20 m. How much shorter is it to cut diagonally across than to walk around two sides, correct to one decimal place?

38. A ramp rises 0.5 m over a horizontal distance of 6 m. The ramp surface is to be covered with non-slip strip costing \$18 per metre. Find the cost, to the nearest dollar.

Conceptual

39. A ladder is moved further from the wall. Explain what happens to the height it reaches.

40. Sam walks 3 km north, then 4 km north-east. He says he is 5 km from his start. Explain why Pythagoras' theorem does not apply.

41. A 4 m ladder must reach 3.9 m up a wall. How far from the wall can its foot be, correct to two decimal places?

42. Explain why the diagonal of a rectangle is always shorter than the sum of its length and width.

43. A plan of a triangular paddock ABC has a scale of $1 : 400$. The angle at B is a right angle.

(a) On the plan, AB measures 15 cm. Find the actual length of AB in metres. (2 marks)

(b) BC is the same length as AB. Find AC, correct to 3 significant figures. (3 marks)

Revision

44. A bag holds 4 green, 6 yellow and 10 blue marbles. One marble is taken at random.

(a) Find $P(\text{yellow})$ in simplest form. (1 mark)

(b) Find the probability that the marble is not blue, as a percentage. (2 marks)

(c) In 60 draws with replacement, how many greens would you expect? (2 marks)

45. A cylindrical can has a diameter of 8 cm and a height of 12 cm.

(a) Find the volume of the can, correct to the nearest cm^3. (2 marks)

(b) Find its capacity in millilitres. (1 mark)

(c) Find the area of the curved surface, to the nearest cm^2. (2 marks)

46. A builder's plan has a scale of 1 : 50. A wall measures 9.6 cm on the plan.

(a) Find the actual length of the wall in metres. (2 marks)

(b) The wall is 2.4 m high. Find its area. (1 mark)

(c) 1 L of paint covers 12 m^2. How many 4 L tins for two coats? (2 marks)

3 The trigonometric ratios

ANCHOR QUESTION

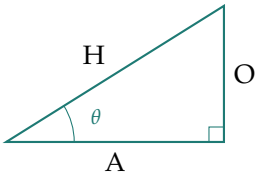
A right-angled triangle has sides 5, 12 and 13. The side opposite θ is 5. Write $\sin\theta$ as a fraction.

SYLLABUS MST-12-S1-04

- Apply trigonometric ratios to find the lengths of unknown sides in right-angled triangles given an angle in both degrees, and degrees and minutes

NOTES

- Name the sides from the marked angle θ. **Hypotenuse (H):** opposite the right angle. **Opposite (O):** across from θ. **Adjacent (A):** next to θ, not the hypotenuse.
- If the marked angle moves to the other acute angle, O and A swap. H never changes.
- $\sin\theta = \dfrac{O}{H}$, $\cos\theta = \dfrac{A}{H}$, $\tan\theta = \dfrac{O}{A}$ (SOH CAH TOA).
- Choose the ratio that uses the two sides in the question: the side you know and the side you want.
- Set the calculator to **degree** mode before finding sin, cos or tan of an angle.



WORKED EXAMPLE

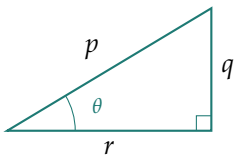
A right-angled triangle has sides 8, 15 and 17. The side opposite θ is 8. Write $\sin\theta$, $\cos\theta$ and $\tan\theta$.

$\mathrm{H} = 17, \mathrm{O} = 8, \mathrm{A} = 15$ — name the sides from θ

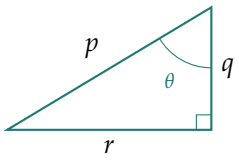
$\sin\theta = \frac{8}{17}$, $\cos\theta = \frac{15}{17}$, $\tan\theta = \frac{8}{15}$ — SOH CAH TOA

Tier 1 · Naming the sides

47. Name the hypotenuse, opposite and adjacent sides for the angle θ.



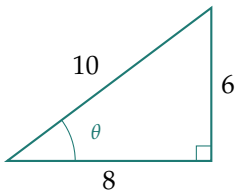
48. Name the hypotenuse, opposite and adjacent sides for the angle θ.



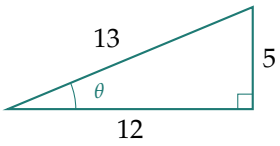
49. In triangle XYZ, the right angle is at Y. For the angle at X, which side is the opposite side?

Tier 2 · Writing one ratio

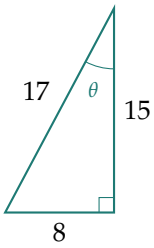
50. Write $\tan\theta$ as a fraction.



51. Write $\cos\theta$ as a fraction.



52. Write $\sin\theta$ as a fraction.



Tier 3 · All three ratios

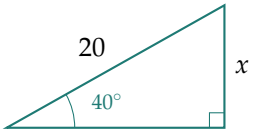
53. A right-angled triangle has sides 3, 4 and 5. The side opposite θ is 3. Write $\sin\theta$, $\cos\theta$ and $\tan\theta$.

54. A right-angled triangle has sides 7, 24 and 25. The side adjacent to θ is 7. Write $\sin\theta$, $\cos\theta$ and $\tan\theta$.

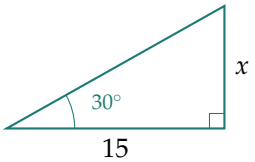
55. A right-angled triangle has sides 20, 21 and 29. The side opposite θ is 20. Write $\sin\theta$, $\cos\theta$ and $\tan\theta$.

Tier 4 · Choosing the ratio

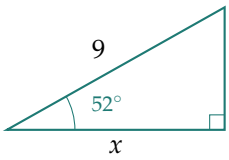
56. Write an equation connecting x and the given side.



57. Write an equation connecting x and the given side.



58. Write an equation connecting x and the given side.



Tier 5 · Using a calculator

59. Find $\sin 50°$, correct to 4 decimal places.

60. Find $\cos 72°$, correct to 4 decimal places.

61. Find $\tan 25°$ and $\tan 65°$, correct to 4 decimal places. Which is greater than 1?

Conceptual

62. Explain why $\sin\theta$ and $\cos\theta$ are always less than 1 for an acute angle in a right-angled triangle.

63. Jake writes $\tan\theta = \frac{A}{O}$. Explain his mistake.

64. In a right-angled triangle, $\sin\theta = \frac{3}{5}$. Find $\cos\theta$ and $\tan\theta$.

65. Explain why the opposite and adjacent sides change when the marked angle moves to the other acute angle.

66. A right-angled triangle has a hypotenuse of 41 and a shorter side of 9, which is opposite the angle θ.

(a) Find the length of the third side. (2 marks)

(b) Write $\tan\theta$ as a fraction. (1 mark)

(c) Which trigonometric ratio of θ is equal to $\frac{40}{41}$? (1 mark)

Revision

67. A ladder 7 m long leans against a wall. Its foot is 1.8 m from the base of the wall.

(a) How high up the wall does the ladder reach, to two decimal places? (2 marks)

(b) Is the height at least 4 times the distance from the wall? Justify. (2 marks)

(c) Write the ratio of distance from the wall to height, in the form $1 : n$. (1 mark)

68. A coin is tossed and a spinner with sections A, B and C is spun.

(a) Draw a tree diagram and list the outcomes. (2 marks)

(b) Find $P(\text{tail and C})$. (1 mark)

(c) Find $P(\text{head or B, or both})$. (2 marks)

69. Priya earns \$1240 a week. Her tax is \$214 a week and the Medicare levy is 2% of her pay.

(a) Find the Medicare levy each week. (1 mark)

(b) Find her net (take-home) pay each week. (2 marks)

(c) Find her net pay for a year (52 weeks). (2 marks)

4 Finding a side: unknown on top

ANCHOR QUESTION

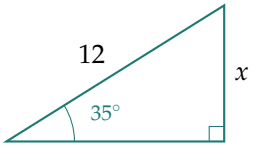
A right-angled triangle has a hypotenuse of 12 and an angle of 35°. Find the side opposite the angle, correct to two decimal places.

SYLLABUS MST-12-S1-04

- Apply trigonometric ratios to find the lengths of unknown sides in right-angled triangles given an angle in both degrees, and degrees and minutes

NOTES

- Label the sides H, O and A from the given angle, then choose the ratio that uses the known side and the unknown side.
- Write the equation, e.g. $\sin 35° = \dfrac{x}{12}$.
- When x is on top, multiply both sides by the bottom number: $x = 12 \sin 35°$.
- Check: the hypotenuse must be the longest side.



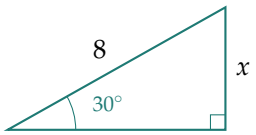
WORKED EXAMPLE

A right-angled triangle has a hypotenuse of 20 and an angle of 25°. Find the side adjacent to the angle, correct to two decimal places.

A and H: cosine	choose the ratio
$\cos 25° = \frac{x}{20}$	CAH
$x = 20 \cos 25° = 18.126\ldots \approx 18.13$	multiply by 20

Tier 1 · Sine

70. Find x, correct to two decimal places.

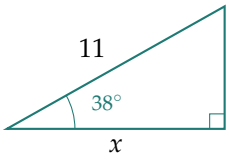


71. A right-angled triangle has a hypotenuse of 15 and an angle of 42°. Find the side opposite the angle, to two decimal places.

72. A right-angled triangle has a hypotenuse of 6.4 and an angle of 67°. Find the side opposite the angle, to two decimal places.

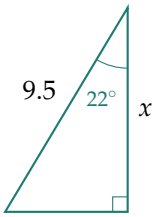
Tier 2 · Cosine

73. Find x, correct to two decimal places.



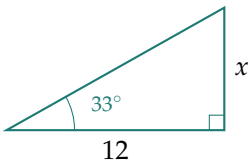
74. A right-angled triangle has a hypotenuse of 24 and an angle of 55°. Find the side adjacent to the angle, to two decimal places.

75. Find x, correct to two decimal places.



Tier 3 · Tangent

76. Find x, correct to two decimal places.

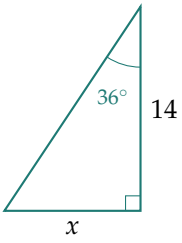


77. A right-angled triangle has a side of 5 adjacent to an angle of 61°. Find the side opposite the angle, to two decimal places.

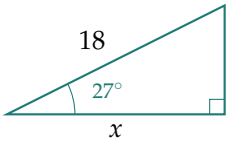
78. A right-angled triangle has a side of 40 adjacent to an angle of 15°. Find the side opposite the angle, to one decimal place.

Tier 4 · Choosing the ratio

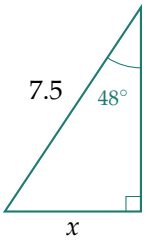
79. Find x, correct to two decimal places.



80. Find x, correct to two decimal places.



81. Find x, correct to two decimal places.



Tier 5 · Practical problems

82. A 7 m ladder leans against a wall, making an angle of 75° with the ground. How high up the wall does it reach, to two decimal places?

83. A kite string is 50 m long and makes an angle of 40° with the ground. How high is the kite, to the nearest metre?

84. A tree casts a shadow 18 m long. The sun's rays make an angle of 34° with the ground. How tall is the tree, to one decimal place?

Conceptual

85. Explain how you decide whether to use sin, cos or tan to find a side.

86. Zoe finds a shorter side as $x = 10\sin 30^\circ = -9.88$. Explain what has gone wrong.

87. In a right-angled triangle with hypotenuse 10, explain why the side opposite 60° is longer than the side opposite 30°.

88. Explain why $\tan 45^\circ = 1$ using a right-angled triangle.

89. A ramp 6.2 m long makes an angle of 6° with the ground.

(a) Find the height of the ramp, to two decimal places. (2 marks)

(b) Find the horizontal distance the ramp covers, to two decimal places. (2 marks)

Revision

90. A rectangular sports field is 105 m long and 68 m wide.

(a) Find the length of the diagonal, to the nearest metre. (2 marks)

(b) Find the area of the field in hectares. (2 marks)

(c) A player jogs around the edge 3 times. How far is this in km? (1 mark)

91. A survey found that 18 of 75 customers bought a coffee with breakfast.

(a) Find the relative frequency of buying a coffee, as a decimal. (1 mark)

(b) Estimate how many of next month's 1200 customers will buy a coffee. (2 marks)

(c) Coffee costs \$5.20. Estimate the coffee sales for the month. (2 marks)

92. A flight leaves Perth at 11:20 pm and lands in Sydney 4 hours 10 minutes later. Sydney is 3 hours ahead of Perth.

(a) What is the time in Perth when the flight lands? (1 mark)

(b) What is the local time in Sydney when the flight lands? (2 marks)

(c) The flight is 3290 km. Find the average speed, to the nearest km/h. (2 marks)

5 Finding a side: unknown on the bottom

ANCHOR QUESTION

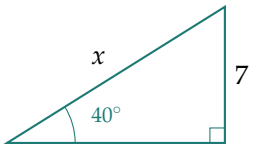
A right-angled triangle has a side of 7 opposite an angle of 40°. Find the hypotenuse, correct to two decimal places.

SYLLABUS MST-12-S1-04

- Apply trigonometric ratios to find the lengths of unknown sides in right-angled triangles given an angle in both degrees, and degrees and minutes

NOTES

- Choose the ratio as before. Sometimes the unknown ends up on the **bottom** of the fraction: $\sin 40° = \dfrac{7}{x}$.
- When x is on the bottom, swap x and the ratio, so you **divide**: $x = \dfrac{7}{\sin 40°}$.
- When x is on top, multiply. Decide which before you press any calculator keys.
- Check: if x is the hypotenuse, it must be longer than the other sides.



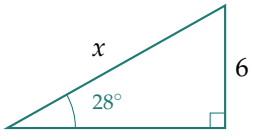
WORKED EXAMPLE

A right-angled triangle has a side of 12 adjacent to an angle of 35°. Find the hypotenuse, correct to two decimal places.

A and H: cosine — choose the ratio
$\cos 35° = \frac{12}{x}$ — x is on the bottom
$x = \frac{12}{\cos 35°} = 14.649\ldots \approx 14.65$ — divide

Tier 1 · Sine

93. Find x, correct to two decimal places.

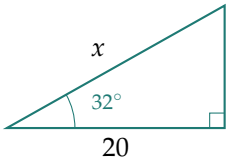


94. A right-angled triangle has a side of 15 opposite an angle of 62°. Find the hypotenuse, to two decimal places.

95. A right-angled triangle has a side of 3.6 opposite an angle of 19°. Find the hypotenuse, to two decimal places.

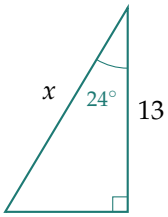
Tier 2 · Cosine

96. Find x, correct to two decimal places.



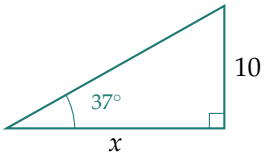
97. A right-angled triangle has a side of 8 adjacent to an angle of 71°. Find the hypotenuse, to two decimal places.

98. Find x, correct to one decimal place.



Tier 3 · Tangent

99. Find x, correct to two decimal places.



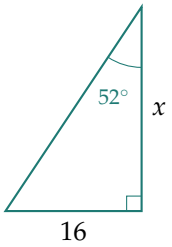
100. A right-angled triangle has a side of 25 opposite an angle of 66°. Find the side adjacent to the angle, to two decimal places.

101. A right-angled triangle has a side of 4.8 opposite an angle of 12°. Find the side adjacent to the angle, to one decimal place.

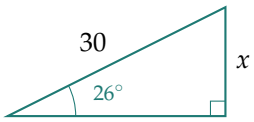
Tier 4 · Top or bottom?

102. In a right-angled triangle, an angle is $44°$. (a) The hypotenuse is 9; find the adjacent side. (b) The adjacent side is 9; find the hypotenuse. Give both to two decimal places.

103. Find x, correct to two decimal places.



104. Find x, correct to two decimal places.



Tier 5 · Practical problems

105. A ladder reaches 4.5 m up a wall and makes an angle of $70°$ with the ground. How long is the ladder, to two decimal places?

106. A path rises 28 m and makes an angle of $6°$ with the horizontal. Find the length of the path, to the nearest metre.

107. A flagpole 9 m tall casts a shadow. The sun's rays make an angle of $55°$ with the ground. Find the length of the shadow, to one decimal place.

Conceptual

108. Explain how you know whether to multiply or divide after writing the trigonometric equation.

109. Liam solves $\sin 30° = \frac{5}{x}$ and gets $x = 5 \sin 30° = 2.5$. Explain his mistake and find x.

110. A right-angled triangle has a side of 10 opposite an angle of $45°$. Without a calculator, find the adjacent side. Explain.

111. Explain why dividing by $\sin\theta$ always gives an answer larger than the side you started with, for an acute angle θ.

112. A kite is flying 32 m above the ground. Its string makes an angle of $46°$ with the ground.

(a) Find the length of the string, to two decimal places. (2 marks)

(b) Find the horizontal distance from the person to the point below the kite. (2 marks)

Revision

113. A 5 m ladder makes an angle of 68° with the ground.

(a) How high up the wall does it reach, to two decimal places? (2 marks)

(b) How far is its foot from the wall, to two decimal places? (2 marks)

(c) Check your answers using Pythagoras' theorem. (1 mark)

114. Two dice are rolled and the numbers are added.

(a) Find $P(\text{total of } 4)$. (1 mark)

(b) Find $P(\text{total is not } 4)$. (1 mark)

(c) The dice are rolled 240 times. How many totals of 4 would you expect? (2 marks)

(d) In 240 rolls, a total of 4 came up 31 times. Find the relative frequency. (1 mark)

115. Liam is paid \$27.60 per hour. Saturday hours are paid at time-and-a-half.

(a) Find his pay for 32 weekday hours. (1 mark)

(b) Find his pay for 6 Saturday hours. (2 marks)

(c) Find his total weekly pay, and his pay for a fortnight. (2 marks)

6 Degrees and minutes

ANCHOR QUESTION

Round 41.2845° *to the nearest minute.*

SYLLABUS MST-12-S1-04

- Apply trigonometric ratios to find the lengths of unknown sides in right-angled triangles given an angle in both degrees, and degrees and minutes

NOTES

- Angles can be measured in degrees and minutes, like hours and minutes: $1^\circ = 60'$.
- $35^\circ 30'$ means 35 degrees and 30 minutes, which is 35.5°.
- Decimal part to minutes: multiply by 60. Minutes to a decimal: divide by 60.
- Round to the nearest minute by rounding the minutes. If they round to $60'$, add 1°.
- On the calculator, the **DMS** key (marked $^\circ\ '\ ''$) enters and converts degrees and minutes. Enter the angle this way inside sin, cos or tan.

WORKED EXAMPLE

Round 56.3726° *to the nearest minute.*

$0.3726 \times 60 = 22.356'$ — decimal part $\times 60$
$\approx 22'$ — round the minutes
$56^\circ 22'$

Tier 1 · Converting

116. Write 48.5° in degrees and minutes.

117. Write 72.4° in degrees and minutes.

118. Write $33^\circ 45'$ as a decimal number of degrees.

Tier 2 · Rounding to the nearest minute

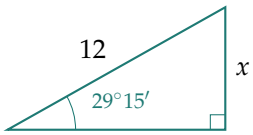
119. Round 19.8462° to the nearest minute.

120. Round 64.1032° to the nearest minute.

121. Round 27.9958° to the nearest minute.

Tier 3 · Sides with degrees and minutes

122. Find x, correct to two decimal places.



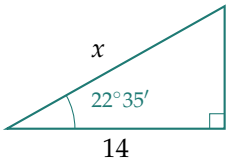
123. A right-angled triangle has a hypotenuse of 20 and an angle of $63°50′$. Find the side adjacent to the angle, to two decimal places.

124. A right-angled triangle has a side of 7 adjacent to an angle of $41°20′$. Find the side opposite the angle, to two decimal places.

Tier 4 · Unknown on the bottom

125. A right-angled triangle has a side of 9 opposite an angle of $36°40′$. Find the hypotenuse, to two decimal places.

126. Find x, correct to two decimal places.



127. A right-angled triangle has a side of 30 opposite an angle of $58°12′$. Find the side adjacent to the angle, to two decimal places.

Tier 5 · Practical problems

128. A 9 m ladder makes an angle of $72°30′$ with the ground. How high up the wall does it reach, to two decimal places?

129. A ramp rises 0.8 m and makes an angle of $5°10′$ with the ground. Find the length of the ramp, to two decimal places.

130. A building casts a shadow 26 m long. The sun's rays make an angle of $47°35′$ with the ground. Find the height of the building, to the nearest metre.

Conceptual

131. Explain why $25.5°$ is $25°30'$ and not $25°50'$.

132. Kim writes $18.9917°$ to the nearest minute as $18°60'$. Explain why this should be written differently.

133. Which angle is larger: $40°45'$ or $40.7°$? Explain.

134. Explain how rounding an angle to the nearest minute is like rounding a time to the nearest minute.

135. A wheelchair ramp rises 0.75 m and makes an angle of $4°30'$ with the ground.

(a) Find the length of the ramp, to two decimal places. (2 marks)

(b) Find the horizontal distance it covers, to two decimal places. (2 marks)

Revision

136. A flight leaves Sydney at 7:35 am and takes 2 hours 55 minutes to reach Auckland, which is 2 hours ahead of Sydney.

(a) Find the arrival time in Sydney time. (1 mark)

(b) Find the local time in Auckland when the flight lands. (1 mark)

(c) The flight is 2160 km. Find the average speed, to the nearest km/h. (2 marks)

(d) Write 2 hours 55 minutes as a decimal, to two decimal places. (1 mark)

137. A right-angled triangle has a hypotenuse of 26 and a shorter side of 10.

(a) Find the third side. (2 marks)

(b) 10 is opposite θ. Write $\sin\theta$ and $\tan\theta$ in simplest form. (2 marks)

(c) Find the perimeter of the triangle. (1 mark)

138. A spinner has 8 equal sections: 3 red, 2 blue and 3 green.

(a) List the sample space. (1 mark)

(b) Find $P(\text{not blue})$. (1 mark)

(c) The spinner is spun twice. Find $P(\text{both blue})$ using an array. (2 marks)

(d) In 120 spins, how many reds would you expect? (1 mark)

7 Finding an angle

ANCHOR QUESTION

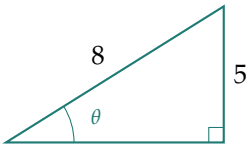
A right-angled triangle has a hypotenuse of 8 and a side of 5 opposite the angle θ. Find θ, correct to the nearest degree.

SYLLABUS MST-12-S1-04

- Apply trigonometric ratios to find the size of unknown angles in right-angled triangles, in both degrees, and degrees and minutes

NOTES

- Name the sides from θ and choose the ratio that uses the two known sides.
- Write the ratio as a fraction, e.g. $\sin\theta = \frac{5}{8}$.
- Use the inverse (SHIFT then sin, cos or tan): $\theta = \sin^{-1}\left(\frac{5}{8}\right)$.
- Round to the nearest degree, or use the DMS key to round to the nearest minute.
- Check: the angle opposite the longer of the two shorter sides is the larger acute angle.



WORKED EXAMPLE

A right-angled triangle has a side of 6 opposite θ and a side of 11 adjacent to θ. Find θ, correct to the nearest minute.

O and A: tangent	choose the ratio
$\tan\theta = \frac{6}{11}$	TOA
$\theta = \tan^{-1}\left(\frac{6}{11}\right) = 28.610\ldots^\circ = 28^\circ 37'$	inverse, then DMS

Tier 1 · From a ratio

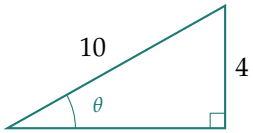
139. Find θ if $\cos\theta = 0.5$.

140. Find θ, to the nearest degree, if $\tan\theta = 2.4$.

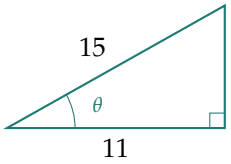
141. Find θ, to the nearest minute, if $\sin\theta = 0.35$.

Tier 2 · Nearest degree

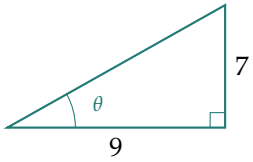
142. Find θ, to the nearest degree.



143. Find θ, to the nearest degree.



144. Find θ, to the nearest degree.



Tier 3 · Nearest minute

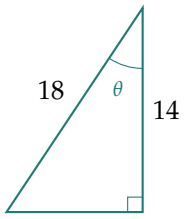
145. A right-angled triangle has a hypotenuse of 13 and a side of 9 opposite θ. Find θ, to the nearest minute.

146. A right-angled triangle has a hypotenuse of 25 and a side of 17 adjacent to θ. Find θ, to the nearest minute.

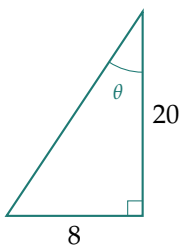
147. A right-angled triangle has a side of 12 opposite θ and 5 adjacent to θ. Find θ, to the nearest minute.

Tier 4 · Choosing the ratio

148. Find θ, to the nearest minute.



149. Find θ, to the nearest minute.



150. In a right-angled triangle, the two shorter sides are 6 and 10. Find the smaller acute angle, to the nearest degree.

Tier 5 · Practical problems

151. A ramp rises 0.9 m over a horizontal distance of 11 m. Find the angle it makes with the ground, to the nearest minute.

152. A 7 m ladder reaches 6.6 m up a wall. Find the angle it makes with the ground, to the nearest degree.

153. A 15 m flagpole casts a 22 m shadow. Find the angle the sun's rays make with the ground, to the nearest minute.

Conceptual

154. Explain why $\sin\theta = 1.2$ has no answer for an angle in a right-angled triangle.

155. Ava finds $\tan\theta = \frac{3}{4}$ and writes $\theta = \tan(0.75) = 0.013$. Explain her mistake.

156. In a right-angled triangle, one acute angle is $37°12'$. Find the other acute angle.

157. Explain how to check an angle found with $\tan^{-1}$ by using the original sides.

158. An 8 m ladder leans against a wall. Its foot is 2 m from the wall.

(a) Find the angle the ladder makes with the ground, to the nearest degree. (2 marks)

(b) Find how high up the wall it reaches, to two decimal places. (2 marks)

Revision

159. A kite string is 60 m long and makes an angle of $52°15'$ with the ground.

(a) Find the height of the kite, to the nearest metre. (2 marks)

(b) Find the horizontal distance to below the kite, to the nearest metre. (2 marks)

(c) Write $52°15'$ as a decimal number of degrees. (1 mark)

160. A school has 1200 students. In a sample of 80, 28 said they play a musical instrument.

(a) Find the relative frequency, as a percentage. (1 mark)

(b) Estimate how many students at the school play an instrument. (2 marks)

(c) Estimate the probability that a student does not play one. (2 marks)

161. A car uses 7.4 L/100 km. Petrol costs \$1.98 per litre.

(a) Find the fuel used for a 350 km trip. (1 mark)

(b) Find the cost of the fuel for this trip, to the nearest cent. (2 marks)

(c) How far can it travel on a full 52 L tank, to the nearest km? (2 marks)

8 Angles of elevation and depression

ANCHOR QUESTION

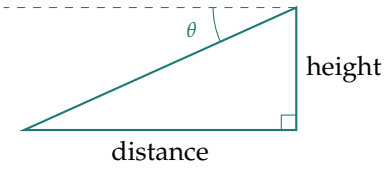
From a point 50 m from the base of a tower, the angle of elevation to the top is $35°$*. Find the height of the tower.*

SYLLABUS MST-12-S1-04

- Solve practical problems involving Pythagoras' theorem, the trigonometry of right-angled triangles, angles of elevation and depression, and true bearings and compass bearings

NOTES

- Angles of elevation and depression are measured from the **horizontal** to the **line of sight**.
- **Angle of elevation:** looking up from the horizontal. **Angle of depression:** looking down from the horizontal.
- The angle of depression from A to B equals the angle of elevation from B to A (alternate angles), so the depression angle can be placed inside the triangle at the lower point.
- Then solve the right-angled triangle as usual: name the sides, choose the ratio, solve.



θ is the angle of depression from the top; it equals the angle of elevation from the bottom.

WORKED EXAMPLE

From the top of an 80 m cliff, the angle of depression to a boat is $20°$*. How far is the boat from the base of the cliff, to the nearest metre?*

Angle at the boat $= 20°$ — alternate angles

$\tan 20° = \frac{80}{x}$ — O and A

$x = \frac{80}{\tan 20°} = 219.8\ldots \approx 220$ m — unknown on the bottom

Tier 1 · Elevation or depression?

162. From the ground, Ana looks up at a kite. Is this an angle of elevation or depression?

163. From a lookout, Ben looks down at a car park. Is this an angle of elevation or depression?

164. The angle of depression from a balcony to a dog is $33°$. What is the angle of elevation from the dog to the balcony? Why?

Tier 2 · Elevation: finding a height

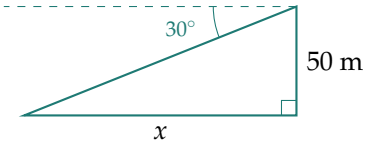
165. From a point 25 m from the base of a tree, the angle of elevation to the top is 38°. Find the height of the tree, to two decimal places.

166. From a point 120 m from the base of a building, the angle of elevation to the top is 27°. Find the height of the building, to the nearest metre.

167. A path to a lookout is 350 m long and rises at an angle of elevation of 6°. How much higher is the lookout than the start of the path, to the nearest metre?

Tier 3 · Depression: finding a distance

168. From the top of a 50 m cliff, the angle of depression to a boat is 30°. How far is the boat from the base of the cliff, to two decimal places?



169. From the top of a 72 m building, the angle of depression to a car is 41°. How far is the car from the base of the building, to the nearest metre?

170. From the top of a 30 m tower, the angle of depression to a cyclist is 15°. Find the line-of-sight distance from the top of the tower to the cyclist, to the nearest metre.

Tier 4 · Finding the angle

171. A 40 m tower is 90 m from Kai. Find the angle of elevation from Kai to the top of the tower, to the nearest degree.

172. A lookout is 150 m above sea level. A ship is 800 m from the base of the lookout. Find the angle of depression to the ship, to the nearest minute.

173. A 2 km straight road climbs 160 m. Find its angle of elevation, to the nearest minute.

Tier 5 · Two-step problems

174. From the top of a 60 m lighthouse, the angle of depression to a boat is 14°. The boat then sails 100 m towards the lighthouse. Find the new angle of depression, to the nearest degree.

175. Mia stands 30 m from a building. The angle of elevation to the top is 52°. Her eyes are 1.6 m above the ground. Find the height of the building, to one decimal place.

176. From the top of a 45 m cliff, the angles of depression to two buoys in a line with the cliff are 40° and 25°. How far apart are the buoys, to the nearest metre?

Conceptual

177. Explain why angles of elevation and depression are measured from the horizontal and not from the vertical.

178. Leo marks a 30° angle of depression between the cliff face and the line of sight. Explain his mistake.

179. As you walk towards a tall building, what happens to the angle of elevation to the top? Explain.

180. Explain why a diagram is useful before solving an elevation or depression problem.

181. In a national park, a straight path connects a car park to a lookout. The lookout is 48 m higher than the car park. The path has an angle of elevation of 5°.

(a) Find the length of the path, to the nearest metre. (2 marks)

(b) Find the horizontal distance covered, to the nearest metre. (2 marks)

Revision

182. A right-angled triangle has shorter sides of 8.4 cm and 11.2 cm.

(a) Find the hypotenuse. (2 marks)

(b) Find the smaller acute angle, to the nearest minute. (2 marks)

(c) Find the area of the triangle. (1 mark)

183. Two coins are tossed 160 times.

(a) List the outcomes for one toss of the two coins. (1 mark)

(b) Find $P(\text{at least one head})$. (2 marks)

(c) How many times would you expect at least one head in 160 tosses? (2 marks)

184. A \$24 000 car is bought with a 15% deposit. The rest is paid over 36 equal monthly payments of \$640.

(a) Find the deposit. (1 mark)

(b) Find the total paid for the car. (2 marks)

(c) How much more than the cash price is this? (2 marks)

9 True and compass bearings

ANCHOR QUESTION

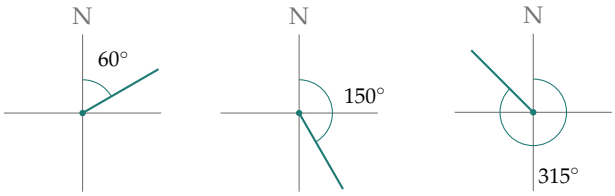
Write 045°*T as a compass bearing.*

SYLLABUS MST-12-S1-04

- Identify and interpret true bearings and compass bearings and convert between them

NOTES

- A **true bearing** is measured clockwise from north and written with three digits: 060°T, 150°T, 315°T.
- A **compass bearing** starts at N or S and turns towards E or W, by an angle from 0° to 90°: N60°E, S30°E, N45°W.
- True to compass: between N and E, Nb°E; between E and S, S$(180 - b)$°E; between S and W, S$(b - 180)$°W; between W and N, N$(360 - b)$°W.
- The bearing of A from B is measured at B. To turn around, add or subtract 180°.



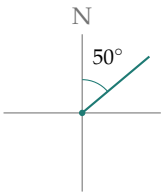
WORKED EXAMPLE

Write 205°*T as a compass bearing.*

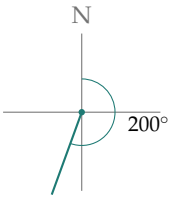
205° is between S (180°) and W (270°) — find the quarter
205° − 180° = 25° — angle from south
S25°W — start at S, turn to W

Tier 1 · Reading true bearings

185. Write the true bearing shown.



186. Write the true bearing shown.



187. What true bearing is due west? Due south?

Tier 2 · True to compass

188. Write 072°T as a compass bearing.

189. Write 115°T as a compass bearing.

190. Write 255°T and 330°T as compass bearings.

Tier 3 · Compass to true

191. Write N38°E as a true bearing.

192. Write S15°E as a true bearing.

193. Write S48°W and N62°W as true bearings.

Tier 4 · Turning around

194. The bearing of B from A is 040°T. Find the bearing of A from B.

195. The bearing of Q from P is 295°T. Find the bearing of P from Q.

196. The bearing of a ship from a lighthouse is S20°E. Find the bearing of the lighthouse from the ship, as a compass bearing and as a true bearing.

Conceptual

197. Explain why true bearings are always written with three digits.

198. Ella writes 135°T as N135°E. Explain her mistake.

199. Explain why the bearing of A from B differs from the bearing of B from A by 180°.

200. Which compass bearing is the same as SW? Write it as a true bearing too.

201. The bearing of town B from town A is 065°T.

(a) Write this bearing as a compass bearing. (1 mark)

(b) Find the true bearing of A from B. (1 mark)

(c) Write your answer to (b) as a compass bearing. (1 mark)

Revision

202. From the top of a 70 m cliff, the angle of depression to a buoy is 18°.

(a) How far is the buoy from the base of the cliff, to the nearest metre? (2 marks)

(b) Find the line-of-sight distance to the buoy, to the nearest metre. (2 marks)

(c) What is the angle of elevation from the buoy to the cliff top? (1 mark)

203. A ramp rises 0.45 m over a horizontal distance of 5.4 m.

(a) Find the angle the ramp makes with the ground, to the nearest minute. (2 marks)

(b) Find the length of the ramp, to two decimal places. (2 marks)

(c) Write the ratio rise : horizontal distance in simplest form. (1 mark)

204. A bag holds 5 red and 3 blue counters. Two are drawn without replacement.

(a) How many ordered outcomes are there? (1 mark)

(b) Find the probability that both are blue. (2 marks)

(c) Find the probability that at least one is red. (2 marks)

10 Bearing problems

ANCHOR QUESTION

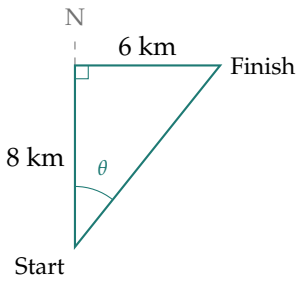
Mia walks 8 km due north, then 6 km due east. How far is she from her start, and what is her true bearing from her start?

SYLLABUS MST-12-S1-04

- Identify and interpret true bearings and compass bearings and convert between them
- Solve practical problems involving Pythagoras' theorem, the trigonometry of right-angled triangles, angles of elevation and depression, and true bearings and compass bearings

NOTES

- Draw the diagram with a north line at the starting point, and mark the right angle and the known lengths.
- Use Pythagoras' theorem for a distance, and $\tan^{-1}$ (or $\sin^{-1}$, $\cos^{-1}$) for an angle.
- Turn the angle into a bearing: an angle east of north is the bearing; from south, use $180° \pm$ the angle; west of north, use $360° -$ the angle.
- "The bearing of B from A" is measured at A. To find the reverse bearing, add or subtract $180°$.
- For a known bearing and distance, the east and north parts are found with sin and cos of the bearing angle.



WORKED EXAMPLE

A boat sails 9 km due south, then 4 km due west. Find its true bearing from its start, to the nearest degree.

$\tan\theta = \frac{4}{9}$ — θ measured from south
$\theta = 23.96\ldots° \approx 24°$ — inverse tan
$\text{Bearing} = 180° + 24° = 204°\text{T}$ — south, then turn west

Tier 1 · North then east

205. Leo walks 5 km due north, then 12 km due east. How far is he from his start?

206. For Leo's walk above, find his true bearing from his start, to the nearest degree.

207. A drone flies 300 m due north, then 120 m due east. Find its distance and true bearing from its start, to the nearest metre and degree.

Tier 2 · Other directions

208. A boat sails 6 km due south, then 3 km due east. Find its true bearing from its start, to the nearest degree.

209. A plane flies 40 km due south, then 25 km due west. Find its true bearing from its start, to the nearest degree.

210. A hiker walks 186 m due north, then 50 m due west. Find the distance from the start (nearest metre) and the true bearing (nearest degree).

Tier 3 · From a bearing and distance

211. A ship sails 30 km on a bearing of 040°T. How far east and how far north of its start is it, to two decimal places?

212. A plane flies 150 km on a bearing of 120°T. How far south of its start is it, to the nearest km?

213. A runner travels 8 km on a bearing of S25°W. How far west of the start is she, to two decimal places?

Tier 4 · Reverse bearings

214. Q is 20 km due east and 15 km due north of P. Find the true bearing of Q from P and of P from Q, to the nearest degree.

215. B is 60 m due east of A. C is 60 m due north of B. Find the true bearing of A from C.

216. A lighthouse is 12 km due south and 5 km due west of a ship. Find the true bearing of the ship from the lighthouse, to the nearest degree.

Conceptual

217. Explain why a north line is drawn at the starting point before finding a bearing.

218. Zac walks 4 km south then 4 km east, and says his bearing from the start is 045°T. Explain his mistake.

219. A ship sails on a bearing of 090°T. Which direction is this, and why is no trigonometry needed to find how far east it travels?

220. Two walkers leave the same point. One walks 3 km north, the other 4 km east. Explain how to find how far apart they are.

221. A ship sails from port P 24 km due east to Q, then 10 km due north to R.

(a) Find the distance PR. (2 marks)

(b) Find the true bearing of R from P, to the nearest degree. (2 marks)

(c) Find the true bearing of P from R. (1 mark)

Revision

222. From the top of a 55 m building, the angle of depression to a car is $28°15'$.

(a) How far is the car from the base of the building, to the nearest metre? (2 marks)

(b) The car drives 40 m closer. Find the new angle of depression. (3 marks)

223. A map has a scale of $1 : 50\,000$. Two towns are 7.2 cm apart on the map, on a bearing of $135°$T.

(a) Find the actual distance between the towns in km. (2 marks)

(b) Write the bearing as a compass bearing. (1 mark)

(c) How far east of the first town is the second town, to two decimal places? (2 marks)

224. In a survey of 250 drivers, 40 had been fined for speeding.

(a) Find the relative frequency of being fined. (1 mark)

(b) Estimate how many of 6000 drivers have been fined. (2 marks)

(c) Explain why an insurer might use this to set prices. (2 marks)

Solutions

1. $\sqrt{9+16} = \sqrt{25} = 5$

2. $\sqrt{25+144} = \sqrt{169} = 13$

3. $\sqrt{49+576} = \sqrt{625} = 25$

4. $\sqrt{16+81} = \sqrt{97} = 9.848\ldots \approx 9.85$

5. $\sqrt{6.25+36} = \sqrt{42.25} = 6.5$

6. $\sqrt{121+196} = \sqrt{317} = 17.80\ldots \approx 17.8$

7. $\sqrt{100-64} = \sqrt{36} = 6$

8. $\sqrt{625-225} = \sqrt{400} = 20$

9. $\sqrt{289-64} = \sqrt{225} = 15$

10. $\sqrt{144-49} = \sqrt{95} = 9.746\ldots \approx 9.75$

11. $\sqrt{72.25-17.64} = \sqrt{54.61} = 7.389\ldots \approx 7.39$

12. $\sqrt{400-182.25} = \sqrt{217.75} = 14.756\ldots \approx 14.76$

13. 0.6 m and 0.8 m. $\sqrt{0.36+0.64} = 1$ m

14. $\sqrt{6.76-1} = \sqrt{5.76} = 2.4$ m

15. 1.5 km and 0.8 km. $\sqrt{2.25+0.64} = \sqrt{2.89} = 1.7$ km

16. $AD = \sqrt{169-25} = 12$. $AC = \sqrt{144+256} = \sqrt{400} = 20$

17. $AD = \sqrt{100-36} = 8$. $AC = \sqrt{64+36} = 10$

18. $\sqrt{49+49} = \sqrt{98} = 9.899\ldots \approx 9.90$ cm

19. $c^2 = a^2 + b^2$, so c^2 is bigger than a^2 and bigger than b^2. That means c is longer than both a and b.

20. She added instead of subtracting. A shorter side cannot be longer than the hypotenuse. $\sqrt{100-36} = 8$.

21. $s^2 + s^2 = 100$, so $s^2 = 50$ and $s = \sqrt{50} = 7.071\ldots \approx 7.07$ cm.

22. The theorem adds and subtracts the squares of the lengths, which only makes sense when they are measured in the same unit.

23. (a) $\sqrt{81+144} = \sqrt{225} = 15$ cm. (b) Arc $= \frac{1}{2} \times \pi \times 15 = 23.56\ldots$ cm. Perimeter $= 9 + 12 + 23.56 = 44.56 \approx 45$ cm.

24. (a) $5 \times 2 = 10$. (b) H2, H4: $\frac{2}{10} = \frac{1}{5}$. (c) $1 - \frac{1}{5} = \frac{4}{5}$. (d) $300 \times \frac{1}{5} = 60$.

25. (a) $\frac{4}{2} \times (6+9) = 30$ m^2. (b) $30 \times 0.1 = 3$ m^3. (c) $3 \times \$65 + \$45 = \$240$.

26. (a) $9000 \div 15 = 600$ min $= 10$ h. (b) $15 \times 60 = 900$ L/h. (c) 9000 L $= 9$ kL. $9 \times \$3.20 = \$28.80$.

27. $\sqrt{12.25 - 1} = \sqrt{11.25} = 3.354\ldots \approx 3.35$ m

28. $\sqrt{16 + 1.44} = \sqrt{17.44} = 4.176\ldots \approx 4.18$ m

29. $\sqrt{100 - 64} = 6$ m

30. $\sqrt{225 + 49} = \sqrt{274} = 16.55\ldots \approx 16.6$ cm

31. $\sqrt{3.24 + 1.44} = \sqrt{4.68} = 2.163\ldots \approx 2.16$ m

32. $\sqrt{4225 - 3249} = \sqrt{976} = 31.2\ldots \approx 31$ inches

33. $\sqrt{360\,000 + 640\,000} = \sqrt{1\,000\,000} = 1000$ m

34. $\sqrt{144 + 25} = 13$ km

35. $\sqrt{52\,900 + 9025} = \sqrt{61\,925} = 248.8\ldots \approx 249$ km *cf. 2024 Std 1 HSC Q14(a)*

36. Two sides $= 170$ m. Diagonal $= \sqrt{14\,400 + 2500} = 130$ m. Saving $= 40$ m.

37. Two sides $= 40$ m. Diagonal $= \sqrt{800} = 28.28\ldots$ m. Saving ≈ 11.7 m.

38. Length $= \sqrt{36 + 0.25} = 6.020\ldots$ m. Cost $= 6.020\ldots \times 18 = \$108.37 \approx \$108$.

39. The ladder (hypotenuse) stays the same length. A longer base means $h^2 = c^2 - b^2$ is smaller, so it reaches lower.

40. North and north-east are not at right angles, so the path does not make a right-angled triangle.

41. $\sqrt{16 - 15.21} = \sqrt{0.79} = 0.888\ldots \approx 0.89$ m

42. The diagonal is a straight line between two corners; walking along two sides is a longer path between the same points.

43. (a) $15 \times 400 = 6000$ cm $= 60$ m. (b) $AC = \sqrt{60^2 + 60^2} = \sqrt{7200} = 84.85\ldots \approx 84.9$ m. *cf. 2025 Std 1 HSC Q20*

44. (a) $\frac{6}{20} = \frac{3}{10}$. (b) $1 - \frac{10}{20} = \frac{1}{2} = 50\%$. (c) $60 \times \frac{4}{20} = 12$.

45. (a) $\pi \times 4^2 \times 12 = 603.18\ldots \approx 603$ cm^3. (b) 603 mL. (c) $2 \times \pi \times 4 \times 12 = 301.59\ldots \approx 302$ cm^2.

46. (a) $9.6 \times 50 = 480$ cm $= 4.8$ m. (b) $4.8 \times 2.4 = 11.52$ m^2. (c) Two coats $= 23.04$ m^2. $23.04 \div 12 = 1.92$ L, so one 4 L tin.

47. H $= p$, O $= q$, A $= r$

48. H $= p$, O $= r$, A $= q$

49. YZ (across from X)

50. $\frac{6}{8} = \frac{3}{4}$

51. $\frac{12}{13}$

52. O = 8 (bottom side), H = 17: $\frac{8}{17}$

53. $\frac{3}{5}$, $\frac{4}{5}$, $\frac{3}{4}$

54. O = 24: $\frac{24}{25}$, $\frac{7}{25}$, $\frac{24}{7}$

55. $\frac{20}{29}$, $\frac{21}{29}$, $\frac{20}{21}$

56. O and H: $\sin 40^\circ = \frac{x}{20}$

57. O and A: $\tan 30^\circ = \frac{x}{15}$

58. A and H: $\cos 52^\circ = \frac{x}{9}$

59. 0.7660

60. 0.3090

61. $\tan 25^\circ = 0.4663$, $\tan 65^\circ = 2.1445$. $\tan 65^\circ$ is greater than 1.

62. Both have the hypotenuse on the bottom, and the hypotenuse is the longest side, so the fraction is less than 1.

63. Tan is opposite over adjacent: $\tan\theta = \frac{O}{A}$ (TOA). He has the fraction upside down.

64. O = 3, H = 5, so A = $\sqrt{25-9} = 4$. $\cos\theta = \frac{4}{5}$, $\tan\theta = \frac{3}{4}$.

65. The opposite side is the one across from the marked angle, so a different angle has a different side across from it. The hypotenuse stays the same.

66. (a) $\sqrt{1681-81} = \sqrt{1600} = 40$. (b) $\tan\theta = \frac{9}{40}$. (c) $\cos\theta$.

67. (a) $\sqrt{49-3.24} = \sqrt{45.76} = 6.764\ldots \approx 6.76$ m. (b) $4 \times 1.8 = 7.2$ m. The height is only 6.76 m, so no. (c) $1.8 : 6.76 \approx 1 : 3.76$.

68. (a) HA, HB, HC, TA, TB, TC. (b) $\frac{1}{6}$. (c) HA, HB, HC, TB: $\frac{4}{6} = \frac{2}{3}$.

69. (a) $0.02 \times \$1240 = \$24.80$. (b) $\$1240 - \$214 - \$24.80 = \$1001.20$. (c) $52 \times \$1001.20 = \$52\,062.40$.

70. $x = 8\sin 30^\circ = 4$

71. $15\sin 42^\circ = 10.036\ldots \approx 10.04$

72. $6.4\sin 67^\circ = 5.891\ldots \approx 5.89$

73. $x = 11\cos 38^\circ = 8.668\ldots \approx 8.67$

74. $24\cos 55^\circ = 13.765\ldots \approx 13.77$

75. From 22° at the top, x is adjacent. $x = 9.5\cos 22^\circ = 8.808\ldots \approx 8.81$

76. $x = 12\tan 33^\circ = 7.792\ldots \approx 7.79$

77. $5\tan 61^\circ = 9.020\ldots \approx 9.02$

78. $40\tan 15^\circ = 10.71\ldots \approx 10.7$

79. From 36° at the top: 14 is A, x is O. $x = 14\tan 36^\circ = 10.171\ldots \approx 10.17$

80. H and A: $x = 18\cos 27^\circ = 16.038\ldots \approx 16.04$

81. From 48° at the top: x is O, 7.5 is H. $x = 7.5\sin 48^\circ = 5.573\ldots \approx 5.57$

82. $7\sin 75^\circ = 6.761\ldots \approx 6.76$ m

83. $50\sin 40^\circ = 32.1\ldots \approx 32$ m

84. $18\tan 34^\circ = 12.14\ldots \approx 12.1$ m

85. Label the sides from the angle. Use the ratio that contains the side you know and the side you want: O and H sine, A and H cosine, O and A tangent.

86. Her calculator is in radian mode. In degree mode $10\sin 30^\circ = 5$. A length cannot be negative.

87. $10\sin 60^\circ = 8.66$ and $10\sin 30^\circ = 5$. The larger angle is opposite the longer side.

88. With a 45° angle, the other acute angle is also 45°, so the opposite and adjacent sides are equal and $\frac{O}{A} = 1$.

89. (a) $6.2\sin 6^\circ = 0.648\ldots \approx 0.65$ m. (b) $6.2\cos 6^\circ = 6.166\ldots \approx 6.17$ m.

90. (a) $\sqrt{105^2 + 68^2} = \sqrt{15\,649} = 125.09\ldots \approx 125$ m. (b) $105 \times 68 = 7140$ m^2 $= 0.714$ ha. (c) $3 \times 2 \times (105 + 68) = 1038$ m $= 1.038$ km.

91. (a) $\frac{18}{75} = 0.24$. (b) $1200 \times 0.24 = 288$. (c) $288 \times \$5.20 = \$1497.60$.

92. (a) 3:30 am. (b) 3:30 am +3 h = 6:30 am. (c) 4 h 10 min $= 4.1\dot{6}$ h. $3290 \div 4.1\dot{6} = 789.6 \approx 790$ km/h.

93. $x = \frac{6}{\sin 28^\circ} = 12.780\ldots \approx 12.78$

94. $\frac{15}{\sin 62^\circ} = 16.988\ldots \approx 16.99$

95. $\frac{3.6}{\sin 19^\circ} = 11.057\ldots \approx 11.06$

96. $x = \frac{20}{\cos 32^\circ} = 23.583\ldots \approx 23.58$

97. $\frac{8}{\cos 71^\circ} = 24.572\ldots \approx 24.57$

98. From 24° at the top, 13 is adjacent. $x = \frac{13}{\cos 24^\circ} = 14.23\ldots \approx 14.2$

99. $\tan 37^\circ = \frac{10}{x}$, $x = \frac{10}{\tan 37^\circ} = 13.270\ldots \approx 13.27$

100. $\frac{25}{\tan 66^\circ} = 11.130\ldots \approx 11.13$

101. $\frac{4.8}{\tan 12^\circ} = 22.58\ldots \approx 22.6$

102. (a) $9\cos 44^\circ = 6.47$ (top). (b) $\frac{9}{\cos 44^\circ} = 12.51$ (bottom).

103. From 52° at the top: 16 is O, x is A. $\tan 52^\circ = \frac{16}{x}$, $x = \frac{16}{\tan 52^\circ} = 12.500\ldots \approx 12.50$

104. x is on top: $x = 30\sin 26^\circ = 13.151\ldots \approx 13.15$

105. $\frac{4.5}{\sin 70^\circ} = 4.788\ldots \approx 4.79$ m

106. $\frac{28}{\sin 6^\circ} = 267.8\ldots \approx 268$ m *cf. 2024 Std 1 HSC Q25*

107. $\frac{9}{\tan 55^\circ} = 6.30\ldots \approx 6.3$ m

108. If x is on top of the fraction, multiply by the bottom number. If x is on the bottom, divide the known side by the ratio.

109. x is on the bottom, so he should divide: $x = \frac{5}{\sin 30^\circ} = 10$. His answer is also shorter than the opposite side, which is impossible for a hypotenuse.

110. 10. With a 45° angle, the triangle is isosceles, so the opposite and adjacent sides are equal.

111. For an acute angle, $\sin\theta$ is less than 1, and dividing by a number less than 1 makes the result larger.

112. (a) $\frac{32}{\sin 46^\circ} = 44.485\ldots \approx 44.49$ m. (b) $\frac{32}{\tan 46^\circ} = 30.902\ldots \approx 30.90$ m.

113. (a) $5\sin 68^\circ = 4.635\ldots \approx 4.64$ m. (b) $5\cos 68^\circ = 1.873\ldots \approx 1.87$ m. (c) $\sqrt{4.635^2 + 1.873^2} = 5.00$ m. ✓

114. (a) $(1,3), (2,2), (3,1)$: $\frac{3}{36} = \frac{1}{12}$. (b) $\frac{11}{12}$. (c) $240 \times \frac{1}{12} = 20$. (d) $\frac{31}{240} = 0.129\ldots \approx 0.13$.

115. (a) $32 \times \$27.60 = \$883.20$. (b) $6 \times 1.5 \times \$27.60 = \$248.40$. (c) $\$883.20 + \$248.40 = \$1131.60$ a week; $\$2263.20$ a fortnight.

116. $48^\circ 30'$

117. $0.4 \times 60 = 24$: $72^\circ 24'$

118. $45 \div 60 = 0.75$: 33.75°

119. $0.8462 \times 60 = 50.77'$: $19^\circ 51'$

120. $0.1032 \times 60 = 6.19'$: $64^\circ 6'$

121. $0.9958 \times 60 = 59.75'$, which rounds to $60'$: $28^\circ 0'$

122. $x = 12\sin 29^\circ 15' = 5.862\ldots \approx 5.86$

123. $20\cos 63^\circ 50' = 8.817\ldots \approx 8.82$

124. $7\tan 41^\circ 20' = 6.157\ldots \approx 6.16$

125. $\frac{9}{\sin 36^\circ 40'} = 15.074\ldots \approx 15.07$

126. $x = \frac{14}{\cos 22^\circ 35'} = 15.164\ldots \approx 15.16$

127. $\frac{30}{\tan 58^\circ 12'} = 18.596\ldots \approx 18.60$

128. $9\sin 72^\circ 30' = 8.583\ldots \approx 8.58$ m

129. $\frac{0.8}{\sin 5°10'} = 8.882\ldots \approx 8.88$ m

130. $26 \tan 47°35' = 28.45\ldots \approx 28$ m

131. There are 60 minutes in a degree, so half a degree is 30 minutes, not 50.

132. $60' = 1°$, so $18°60'$ is $19°0'$. Minutes are always less than 60.

133. $40°45' = 40.75°$, which is larger than $40.7°$.

134. Both use 60 minutes in one unit (a degree or an hour), so the decimal part is multiplied by 60 and then rounded.

135. (a) $\frac{0.75}{\sin 4°30'} = 9.557\ldots \approx 9.56$ m. (b) $\frac{0.75}{\tan 4°30'} = 9.528\ldots \approx 9.53$ m.

136. (a) 10:30 am. (b) 12:30 pm. (c) $2160 \div 2.91\dot{6} = 740.57\ldots \approx 741$ km/h. (d) 2.92 h.

137. (a) $\sqrt{676 - 100} = 24$. (b) $\sin\theta = \frac{10}{26} = \frac{5}{13}$, $\tan\theta = \frac{10}{24} = \frac{5}{12}$. (c) $26 + 24 + 10 = 60$.

138. (a) $S = \{R, B, G\}$. (b) $1 - \frac{2}{8} = \frac{3}{4}$. (c) $8 \times 8 = 64$ cells, $2 \times 2 = 4$ both blue: $\frac{4}{64} = \frac{1}{16}$. (d) $120 \times \frac{3}{8} = 45$.

139. $\cos^{-1}(0.5) = 60°$

140. $\tan^{-1}(2.4) = 67.38\ldots° \approx 67°$

141. $\sin^{-1}(0.35) = 20.487\ldots° = 20°29'$

142. $\sin\theta = \frac{4}{10}$, $\theta = 23.57\ldots° \approx 24°$

143. $\cos\theta = \frac{11}{15}$, $\theta = 42.83\ldots° \approx 43°$

144. $\tan\theta = \frac{7}{9}$, $\theta = 37.87\ldots° \approx 38°$

145. $\sin\theta = \frac{9}{13}$, $\theta = 43.81\ldots° = 43°49'$

146. $\cos\theta = \frac{17}{25}$, $\theta = 47.15\ldots° = 47°9'$

147. $\tan\theta = \frac{12}{5}$, $\theta = 67.38\ldots° = 67°23'$

148. From θ at the top: 14 is A, 18 is H. $\cos\theta = \frac{14}{18}$, $\theta = 38.94\ldots° = 38°57'$

149. From θ at the top: 8 is O, 20 is A. $\tan\theta = \frac{8}{20}$, $\theta = 21.80\ldots° = 21°48'$

150. The smaller angle is opposite the shorter side: $\tan\theta = \frac{6}{10}$, $\theta = 30.96\ldots° \approx 31°$

151. $\tan\theta = \frac{0.9}{11}$, $\theta = 4.677\ldots° = 4°41'$

152. $\sin\theta = \frac{6.6}{7}$, $\theta = 70.53\ldots° \approx 71°$

153. $\tan\theta = \frac{15}{22}$, $\theta = 34.28\ldots° = 34°17'$

154. $\sin\theta = \frac{O}{H}$, and the hypotenuse is the longest side, so the ratio cannot be more than 1.

155. She used tan instead of the inverse. $\theta = \tan^{-1}(0.75) = 36.87°$.

156. The acute angles add to $90°$: $90° - 37°12' = 52°48'$.

157. Find tan of the answer and compare with $\frac{O}{A}$; they should be equal (allowing for rounding).

158. (a) $\cos\theta = \frac{2}{8}$, $\theta = 75.52\ldots° \approx 76°$. (b) $\sqrt{64-4} = \sqrt{60} = 7.745\ldots \approx 7.75$ m.

159. (a) $60\sin 52°15' = 47.46\ldots \approx 47$ m. (b) $60\cos 52°15' = 36.74\ldots \approx 37$ m. (c) $52.25°$.

160. (a) $\frac{28}{80} = 0.35 = 35\%$. (b) $1200 \times 0.35 = 420$. (c) $1 - 0.35 = 0.65$.

161. (a) $7.4 \times 3.5 = 25.9$ L. (b) $25.9 \times \$1.98 = \$51.28$. (c) $52 \div 7.4 \times 100 = 702.7\ldots \approx 703$ km.

162. Elevation

163. Depression

164. $33°$. They are alternate angles between the two horizontal lines.

165. $25\tan 38° = 19.53$ m

166. $120\tan 27° = 61.1\ldots \approx 61$ m

167. $350\sin 6° = 36.6\ldots \approx 37$ m

168. $\frac{50}{\tan 30°} = 86.60$ m

169. $\frac{72}{\tan 41°} = 82.8\ldots \approx 83$ m

170. $\frac{30}{\sin 15°} = 115.9\ldots \approx 116$ m

171. $\tan^{-1}\left(\frac{40}{90}\right) = 23.96\ldots° \approx 24°$

172. $\tan^{-1}\left(\frac{150}{800}\right) = 10.619\ldots° = 10°37'$

173. $\sin^{-1}\left(\frac{160}{2000}\right) = 4.588\ldots° = 4°35'$

174. First distance $= \frac{60}{\tan 14°} = 240.6$ m. New distance $= 140.6$ m. $\tan^{-1}\left(\frac{60}{140.6}\right) = 23.1\ldots° \approx 23°$.

175. $30\tan 52° = 38.40$ m above eye level. Height $= 38.40 + 1.6 = 40.0$ m.

176. Near: $\frac{45}{\tan 40°} = 53.63$ m. Far: $\frac{45}{\tan 25°} = 96.50$ m. Apart $= 42.9 \approx 43$ m.

177. They describe how far the line of sight is tilted up or down from looking straight ahead, which is horizontal.

178. It must be measured from the horizontal, not the vertical cliff face. His angle is the complement, $60°$.

179. It increases. The height stays the same while the horizontal distance gets smaller, so $\tan\theta = \frac{\text{height}}{\text{distance}}$ gets larger.

180. It shows where the right angle is, which sides are known, and where the angle sits, so the right ratio can be chosen.

181. (a) $\frac{48}{\sin 5^\circ} = 550.7\ldots \approx 551$ m. (b) $\frac{48}{\tan 5^\circ} = 548.6\ldots \approx 549$ m. *cf. 2024 Std 1 HSC Q25*

182. (a) $\sqrt{70.56 + 125.44} = \sqrt{196} = 14$ cm. (b) $\tan^{-1}\left(\frac{8.4}{11.2}\right) = 36.869\ldots^\circ = 36^\circ 52'$. (c) $\frac{1}{2} \times 8.4 \times 11.2 = 47.04$ cm^2.

183. (a) HH, HT, TH, TT. (b) $\frac{3}{4}$. (c) $160 \times \frac{3}{4} = 120$.

184. (a) $0.15 \times \$24\,000 = \$3600$. (b) $\$3600 + 36 \times \$640 = \$3600 + \$23\,040 = \$26\,640$. (c) $\$26\,640 - \$24\,000 = \$2640$.

185. 050°T

186. 200°T

187. 270°T; 180°T

188. N72°E

189. $180 - 115 = 65$: S65°E

190. S75°W; N30°W

191. 038°T

192. $180 - 15 = 165$°T

193. 228°T; 298°T

194. $040 + 180 = 220$°T

195. $295 - 180 = 115$°T

196. N20°W, 340°T

197. So every bearing is clear and the same length, e.g. 045°T cannot be misread, and bearings up to 359° are written the same way.

198. A compass bearing angle is at most 90°. 135°T is between E and S, so it is S45°E.

199. They point in opposite directions along the same line, and opposite directions differ by a half turn.

200. S45°W, 225°T

201. (a) N65°E. (b) 245°T. (c) S65°W.

202. (a) $\frac{70}{\tan 18^\circ} = 215.4\ldots \approx 215$ m. (b) $\frac{70}{\sin 18^\circ} = 226.5\ldots \approx 227$ m. (c) 18° (alternate angles).

203. (a) $\tan^{-1}\left(\frac{0.45}{5.4}\right) = 4.763\ldots^\circ = 4^\circ 46'$. (b) $\sqrt{0.2025 + 29.16} = 5.418\ldots \approx 5.42$ m. (c) $0.45 : 5.4 = 1 : 12$.

204. (a) $8 \times 7 = 56$. (b) $3 \times 2 = 6$: $\frac{6}{56} = \frac{3}{28}$. (c) $1 - \frac{3}{28} = \frac{25}{28}$.

205. $\sqrt{25 + 144} = 13$ km

206. $\tan^{-1}\left(\frac{12}{5}\right) = 67.38\ldots^\circ$: 067°T

207. $\sqrt{90\,000 + 14\,400} = 323.1\ldots \approx 323$ m. $\tan^{-1}\left(\frac{120}{300}\right) = 21.8^\circ$: 022°T

208. $\tan^{-1}\left(\frac{3}{6}\right) = 26.57^\circ$ from south. $180 - 27 = 153$°T

209. $\tan^{-1}\left(\frac{25}{40}\right) = 32.0^\circ$ from south. $180 + 32 = 212$°T

210. $\sqrt{34\,596 + 2500} = 192.6\ldots \approx 193$ m. $\tan^{-1}\left(\frac{50}{186}\right) = 15.05^\circ$ west of north: $360 - 15 = 345$°T *cf. 2024 Std 1 HSC Q14*

211. East $= 30\sin 40^\circ = 19.28$ km. North $= 30\cos 40^\circ = 22.98$ km.

212. Angle from south $= 60^\circ$. South $= 150\cos 60^\circ = 75$ km.

213. West $= 8\sin 25^\circ = 3.38$ km.

214. $\tan^{-1}\left(\frac{20}{15}\right) = 53.1^\circ$: 053°T. Reverse: 233°T.

215. At C, $\tan\theta = \frac{60}{60}$, so $\theta = 45^\circ$ west of south: 225°T. *cf. 2025 Std 1 HSC Q20(c)*

216. From the lighthouse the ship is north and east: $\tan^{-1}\left(\frac{5}{12}\right) = 22.6^\circ$: 023°T.

217. True bearings are measured clockwise from north at the point the bearing is taken from, so north must be marked there.

218. The angle is 45° from south, not north. The bearing is $180 - 45 = 135$°T.

219. Due east. All of the distance travelled is east, so the east distance equals the distance sailed.

220. Their paths are at right angles, so the distance apart is the hypotenuse: $\sqrt{9 + 16} = 5$ km.

221. (a) $\sqrt{576 + 100} = 26$ km. (b) $\tan^{-1}\left(\frac{24}{10}\right) = 67.4^\circ$: 067°T. (c) $067 + 180 = 247$°T.

222. (a) $\frac{55}{\tan 28^\circ 15'} = 102.3\ldots \approx 102$ m. (b) $102.3 - 40 = 62.3$ m. $\tan^{-1}\left(\frac{55}{62.3}\right) = 41.4\ldots^\circ$, about 41°.

223. (a) $7.2 \times 50\,000 = 360\,000$ cm $= 3.6$ km. (b) S45°E. (c) $3.6\sin 45^\circ = 2.55$ km.

224. (a) $\frac{40}{250} = 0.16$. (b) $6000 \times 0.16 = 960$. (c) Drivers who have been fined may be more likely to make claims, so the expected cost of insuring them is higher.